7. SIGNAL RECONSTRUCTION AND SPECTRAL ANALYSIS

Objective

This experiment is divided into four simulations:

1. Reconstruction of Signal from Samples (Ideal Interpolation)
2. Fourier Series of Half-Wave Rectifier Output
3. Fourier Transform of a Continuous-Time Signal
4. Fourier Transform of a Rectangular Waveform

Experiment 7.1

Reconstruction

of Signal from Samples (Ideal Interpolation)

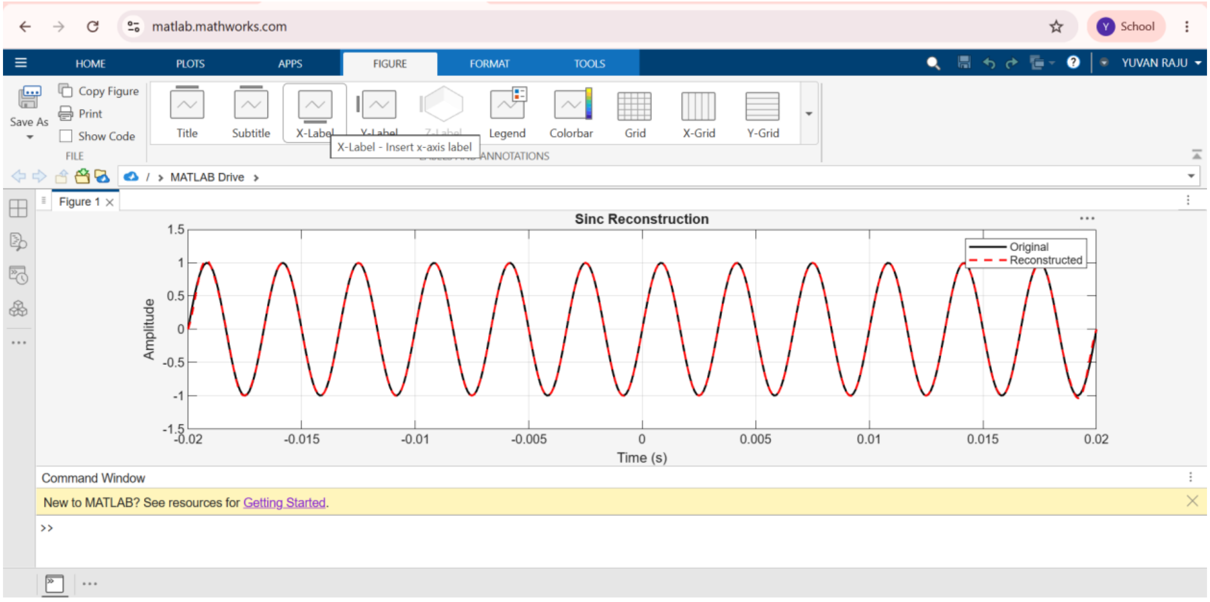
PROGRAM

```
function y = sinc_interp(x, t, Ts)
y = 0*t;
for k = 1:length(x)
y = y + x(k) * sinc((t - (k-1 - length(x)/2)*Ts) / Ts);
end
endfunction
t = -0.02:1e-4:0.02;
f0 = 300;

x = sin(2*%pi*f0*t);
fs = 2000; Ts = 1/fs;
n = -40:40;
ns = n*Ts;
xs = sin(2*%pi*f0*ns);
x_rec = sinc_interp(xs, t, Ts);
clf;
plot(t, x, 'k');
plot(t, x_rec, 'r--');
legend("Original","Reconstructed");
```

xtitle("Sinc Reconstruction");

OUTPUT



Experiment 7.2

Fourier Series of Half-Wave Rectifier Output

PROGRAM

```
% Fourier Series of Half-Wave Rectifier Output
clc;
clear;
close all;

% Define parameters
A = 1;              % Amplitude of input sinusoidal signal
f = 1;             % Frequency (Hz)
T = 1/f;            % Period

% Time vector for two periods
dt = 0.01;
t = 0:dt:2*T;

% Define input sinusoidal signal
```

```
x_t = A * sin(2*pi*f*t);

% Define half-wave rectified output signal
y_t = max(x_t, 0);

% Number of harmonics
N_harmonics = 10;

% Fundamental angular frequency
w = 2*pi*f;

% Initialize Fourier coefficients
a0 = 0;
an = zeros(1, N_harmonics);
bn = zeros(1, N_harmonics);

% Compute Fourier series coefficients
a0 = (2/T) * trapz(t, y_t);

for n = 1:N_harmonics
   an(n) = (2/T) * trapz(t, y_t .* cos(n*w*t));
   bn(n) = (2/T) * trapz(t, y_t .* sin(n*w*t));
end

% Reconstruct signal using Fourier series
y_reconstructed = (a0/2) * ones(size(t));

for n = 1:N_harmonics
   y_reconstructed = y_reconstructed ...
      + an(n)*cos(n*w*t) ...
      + bn(n)*sin(n*w*t);
```

```
end

% Plot the original half-wave rectified signal
figure;

subplot(3,1,1);
plot(t, y_t, 'LineWidth', 1.5);
grid on;
xlabel('t (Time)');
ylabel('Amplitude');
title('Half-Wave Rectified Output Signal');

% Plot reconstructed signal
subplot(3,1,2);
plot(t, y_reconstructed, 'LineWidth', 1.5);
grid on;
xlabel('t (Time)');
ylabel('Amplitude');
title(['Reconstructed Signal Using Fourier Series with ', ...
    num2str(N_harmonics), ' Harmonics']);

% Plot comparison
subplot(3,1,3);
plot(t, y_t, 'LineWidth', 1.5);
hold on;
plot(t, y_reconstructed, '--', 'LineWidth', 1.5);
grid on;
xlabel('t (Time)');
ylabel('Amplitude');
title('Comparison of Original and Reconstructed Signal');
legend('Original Signal', 'Reconstructed Signal');
```

hold off;

OUTPUT